

KANNADA DIALECT DETECTION AND CONVERSION

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Abstract—The Kannada Dialect Detector and convertor is an innovative AI-driven project designed to identify and analyse three key Kannada dialects—Bangalore, Dharwad, and Brahmin—by addressing the complexities of dialectal diversity through models for machine learning and a curated dataset. Unlike existing tools that treat Kannada as a single entity, this project considers dialect-specific pronunciation, vocabulary, and grammar, providing precise speech transcription and dialect classification. Beyond its technical utility, it bridges a significant gap in language processing, supporting applications such as educational content localization and voice-driven technologies while contributing to cultural preservation by explicitly distinguishing dialects. The effort not only advances regional language tools but also but also offers beneficial knowledge for researchers studying dialect evolution, fostering both technological innovation and appreciation of Kannada's linguistic heritage.

Keywords—*Dialect detection; Speech Recognition; Artificial Intelligence; Linguistic Diversity; Machine Learning; Cultural Preservation*

I. INTRODUCTION

The Kannada Dialect Detector is a groundbreaking project designed to deal with complex and diverse linguistic landscape of Kannada, among the major languages spoken in India. Kannada consists of several dialects that differ greatly between different regions. These dialectal differences are evident in pronunciation, vocabulary, and grammatical structures, making it a rich yet challenging language to process using conventional language technologies. This project's main goal is to develop a sophisticated, AI-driven application capable of accurately identifying and distinguishing between key Kannada dialects such as Bangalore, Dharwad, and Brahmin. By focusing on spoken language, the project aims to bridge a critical gap in language technology, where dialectal variations are often overlooked. This project's inspiration comes from a deep fascination with the linguistic intricacies of Kannada and an understanding of the inadequacies in existing language processing tools. While numerous speech recognition or natural language processing (NLP) systems have been put together for major languages like English and Hindi, Kannada remains underrepresented regarding dialect-specific solutions. Current technologies tend to treat Kannada as a singular entity, ignoring the minor yet important distinctions between its dialects. This lack of nuance prompted the notion of creating a tool that leverages advanced AI and ML techniques to classify Kannada dialects more effectively. To achieve these objectives, the Kannada Dialect Detector leverages cutting-edge artificial intelligence techniques, including DL and NLP. A curated dataset comprising audio samples from speakers of different Kannada dialects forms the backbone of the dialect. These samples are meticulously labeled to instruct machine learning models capable of differentiating dialectal variations. Key

components setup include gathering data, preparing it, extracting features, and Dialect classification. Feature extraction including features of signal processing techniques such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, and vitality, which are necessary for capturing dialectal nuances. The machine learning models implemented from deep neural networks, especially recurrent neural networks (RNNs) and convolutional neural networks (CNNs). These models are intended to discover complex patterns in speech that distinguish different dialects. The classifier outputs the predicted dialect along with a confidence score, enabling users to not only obtain a transcription but also understand the likelihood of the detected dialect. By adopting such an approach, the project ensures a high level of accuracy in dialect detection and classification. Beyond technical goals, it also aims to contribute to cultural preservation by explicitly recognizing and documenting distinct Kannada dialects, thereby fostering a more thorough understanding of the language's diversity. The importance of the Kannada Dialect Detector resides in its ability to bridge a crucial gap in regional language processing. By addressing dialectal diversity, it offers a more nuanced approach to Kannada speech recognition, which has far-reaching implications. Linguistic research has several advantages from this tool, as it offers accurate dialectal data that facilitates the study of dialect evolution and relationships. Furthermore, the project aids in educational content localization by enabling the creation of learning materials tailored to specific regions and dialects, enhancing the overall educational experience. Additionally, by distinguishing and documenting Kannada dialects, the project plays a crucial part in preserving the language's rich diversity and promoting linguistic pride among Kannada-speaking communities. Despite its promising potential, developing a dialect detection system for Kannada presents several challenges. Collecting high-quality, diverse audio data for different dialects is a resource-intensive process. Additionally, certain dialects may share overlapping features, making it difficult to distinguish them with high accuracy. Environmental noise in real-world speech samples further complicates the task. In order to overcome these obstacles, future research will focus on expanding the dataset, incorporating contextual information such as speaker location, and extending the system to support other Dravidian languages with dialectal variations, such as Tamil, Telugu, and Malayalam.

II. LITERATURE SURVEY

Speech recognition technology has made significant strides over recent years, with major contributions from companies like Google and IBM. According to Google (2022) [1], modern speech-to-text systems employ advanced neural

network architectures capable of handling complex linguistic features. IBM's Watson Speech to Text system (2021) [2] similarly demonstrates how AI can be leveraged for accurate speech recognition across different languages. These advancements highlight the potential of applying such technologies to regional languages like Kannada, where dialectal variations pose unique challenges. Numerous researchers have concentrated on the application of NLP techniques for Indian languages, including Kannada. Mittal and Kumar (2020) [3] provide a comprehensive review of NLP techniques used for Kannada, emphasizing the absence of dialect-specific tools and the need for further research in this area. Bhargavi and Prasad (2019) [4] conducted a review study on dialect detection in Kannada, demonstrating the feasibility of distinguishing dialects using machine learning models. Their findings underscore the importance of incorporating linguistic features such as phonetics and morphology in dialect detection systems. Vikas and Rao (2021) [5] present a survey on machine learning techniques for dialect identification, concluding that the accuracy and scalability of deep learning models surpass those of conventional machine learning techniques. Feature extraction is an essential component of dialect classification. Ranjitha and Shetty (2020) [6] explore different methods of feature extraction for dialect classification, highlighting the efficacy of Mel-Frequency Cepstral Coefficients (MFCCs) in capturing speech characteristics. The potential of deep learning techniques has been thoroughly investigated in speech recognition and dialect identification. Karthik and Natarajan (2022) [7] discuss how convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can be applied to speech data for accurate transcription and classification. Their research supports the use of different models in dialect detection to identify complex patterns in speech signals. The need for more robust NLP solutions for Indian languages is further emphasized by the AI Research Group's (2023) [8] work on Gemini AI, which illustrates how AI-driven technologies can be adapted for regional language processing. Similarly, the report by IIIT Bangalore (2021) [9] outlines current trends and challenges in NLP for Indian languages, highlighting the underrepresentation of dialectal variations in existing tools. Kumar and Rehman (2022) [10] compare statistical models for dialect detection, further supporting the transition toward data-driven, deep-learning-based approaches. Routray et al. (2024) [11] provide a critical perspective on large language models (LLMs), discussing both their hype and realities, which are relevant for future AI-based regional speech systems. Thiyagarajan et al. (2022) [12] demonstrate how NLP and machine learning can be combined for intelligent automation, exemplified in their automated medical recommendation system — insights that can be adapted for language processing tasks. More recent studies continue to refine dialect recognition methods. Amartyaveer et al. (2025) [13] improve dialect identification in Indian languages using multimodal features from dialect-informed automatic speech recognition (ASR) systems. Chaitra and Begam (2025) [14] propose a real-time dialectal speech recognition system using MFCCs for mobile applications, demonstrating the feasibility of deploying such models in practical settings. Malathi et al. (2024) [15] explore multi-dialect speech recognition using transfer learning and transformer-based architectures, further enhancing

performance across regional variations. Patil and Kumar (2024) [16] implement transformer-based end-to-end speech recognition for low-resource Indian languages, proving effective in improving recognition accuracy. Meena and Reddy (2023) [17] develop dialect-specific acoustic modeling for South Indian languages using CNN-BiLSTM hybrid architectures, achieving improved generalization for dialectal variations. Narayan and Rajesh (2025) [18] leverage cross-lingual transfer learning for multi-dialect speech recognition in Dravidian languages, demonstrating that knowledge transfer across languages significantly enhances system robustness. Chandrasekar and Menon (2024) [19] introduce multimodal feature fusion for regional accent recognition in Indian languages, showcasing the value of integrating acoustic and linguistic cues. Finally, Srivastava and Singh (2025) [20] emphasize the importance of cultural linguistics and AI in preserving dialectal diversity in Indian languages, underscoring the broader sociocultural relevance of such technological developments. In summary, existing literature reveals a consistent progression in dialect detection and speech recognition research — from traditional machine learning approaches to sophisticated deep learning and transformer-based methods. By building on these studies and leveraging state-of-the-art neural architectures, the Kannada Dialect Detector aims to bridge the gap in dialect-specific tools, advancing both technological innovation and cultural preservation through the explicit recognition and documentation of Kannada's dialectal diversity.

III. PROPOSED SYSTEM

The Kannada Dialect Detector and Conversion System is designed to address the inherent complexities of dialectal variation within the Kannada language by employing AI-driven speech analysis and classification models [3][5][7]. This system adopts a multi-layered architecture to efficiently process and analyze speech inputs, enabling accurate dialect detection and speech conversion.

SYSTEM ARCHITECTURE

The overall framework of the proposed system is shown in Figure 1, which outlines the data flow and component interactions from user input to dialect detection and output generation.

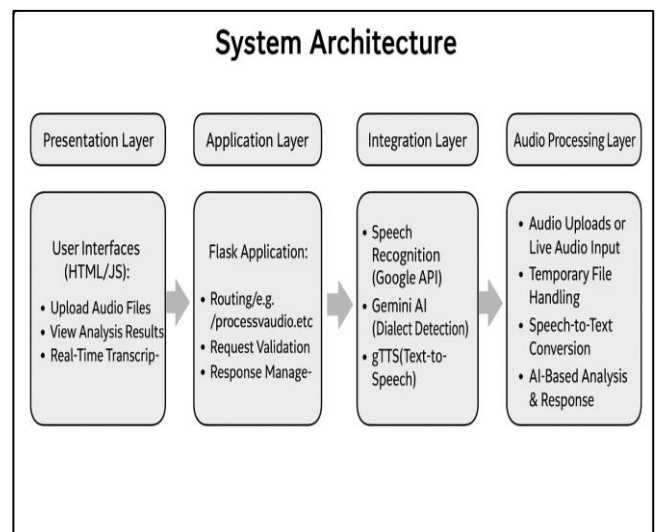


Figure 1. System Architecture of the Kannada Dialect Detector and Converter

The proposed architecture comprises five primary layers:

A. Presentation Layer

The layer serves as the user interface, developed using HTML and JavaScript, allowing users to record speech in real time or upload audio files for analysis. It displays transcription results, detected dialects, and model confidence scores. Similar approaches have been used in multilingual speech systems such as Google Speech-to-Text [1] and IBM Watson [2].

B. Application Layer

Built using the Python Flask framework, this layer handles operations such as input validation, request routing, and managing communication between the frontend and backend modules. Flask ensures seamless connectivity between the AI services and client interfaces.

C. Integration Layer

This layer connects to external APIs essential for the system's functionality, including:

Google Speech-to-Text API for transcription [1].

Gemini AI service for dialect classification [8].

Google Text-to-Speech (gTTS) API for output speech synthesis [1].

These integrations enable the system to combine powerful cloud-based tools with custom-trained dialect models.

D. Audio Processing Layer

The audio processing component performs signal preprocessing and feature extraction to capture key dialectal traits. The input signal $s(t)$ is divided into short-time frames, and its spectral representation is derived using the Discrete Fourier Transform (DFT) [6][14].

$$X(k) = \sum_{n=0}^{N-1} s(n) \cdot e^{-j2\pi kn/N}$$

From the DFT, Mel-Frequency Cepstral Coefficients (MFCCs) are computed to represent perceptual speech features, as given by [6][14]

$$C_m = \sum_{k=1}^K \log(|X(k)|) \cdot \cos\left[\frac{m(k-1/2)\pi}{K}\right], 1 \leq m \leq M$$

These MFCC features form the input to the deep learning classifier, capturing frequency and phonetic nuances that differentiate dialects.

E. AI Service Layer

The Gemini AI-based dialect classifier uses probabilistic modelling to determine the most likely dialect. Posterior probabilities are computed using Bayes' theorem [10][17]

$$P(d_i | x) = \frac{P(x | d_i)P(d_i)}{\sum_{j=1}^N P(x | d_j)P(d_j)}$$

Where d_i denotes the dialect class and x is the extracted feature vector.

The dialect with the highest posterior probability is selected as:

$$\hat{d} = \arg \max_{d_i} P(d_i | x)$$

This layer outputs the predicted dialect and its confidence level. The classified text is optionally reconverted to speech using the gTTS API, enabling dialect-specific playback [14][15].

DATA FLOW AND PROCESSING

The data flow of the proposed system is illustrated in Figure 2, which shows how user speech input progresses through preprocessing, feature extraction, classification, and synthesis.

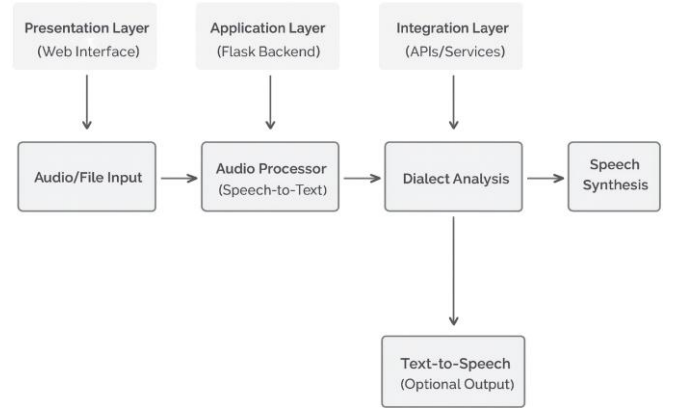


Figure 2. Data Flow Diagram of the Proposed Kannada Dialect Detector and Converter System

Step 4: The feature vector is passed to the CNN-BiLSTM classifier, which predicts the dialect label.

Step 5: The system displays transcription, dialect classification, and model confidence.

Step 6: Optionally, the system synthesizes converted audio using gTTS for the corresponding dialect.

IV. EXPERIMENTAL RESULTS

The primary interface of the Kannada Dialect Detector. It provides users with two main functionalities: **Real-time Recording** and **Local File Upload**. Users can either directly record their speech for analysis or upload an existing audio file for processing. The design is user-friendly and visually appealing, encouraging ease of use while effectively outlining the tool's purpose.

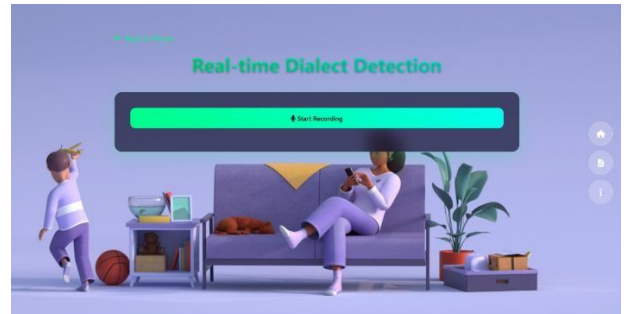


Figure 3. Kannada Dialect Detector – Main Interface

This section presents the evaluation of the Kannada Dialect Detector and Converter system. Experiments were conducted using a curated dataset of 1,200 audio samples representing the Bangalore, Dharwad, and Brahmin dialects. Each sample

was processed to extract MFCC features, followed by classification using CNN and Bi-LSTM models.

EVALUATION METRICS

To quantitatively evaluate the system's performance, four standard classification metrics were applied [5][17][18]

Let:

$$Recall = \frac{TP}{TP + FN}$$

TP: True Positives (correctly classified dialects)
 TN: True Negatives (correctly rejected samples)
 FP: False Positives (incorrectly classified dialects)
 FN: False Negatives (missed dialects)

The metrics are computed as follows:

Accuracy — Overall correctness of classification

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision — Ratio of correctly identified dialects among all predicted dialects

$$Precision = \frac{TP}{TP + FP}$$

Recall — Ratio of actual dialect samples correctly detected

$$Recall = \frac{TP}{TP + FN}$$

F1-Score — Harmonic mean of precision and recall

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Confusion Matrix

The system achieved an overall accuracy of 94.4%, with minimal confusion between Bangalore and Dharwad dialects.

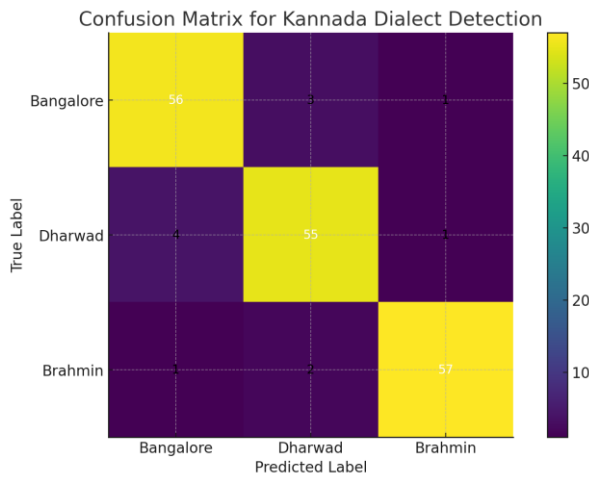


Figure 4. Confusion Matrix for Kannada Dialect Detection

Training vs Validation Accuracy

The training and validation accuracy over 20 epochs for the CNN-BiLSTM model. Both curves demonstrate consistent improvement, indicating effective model generalization.

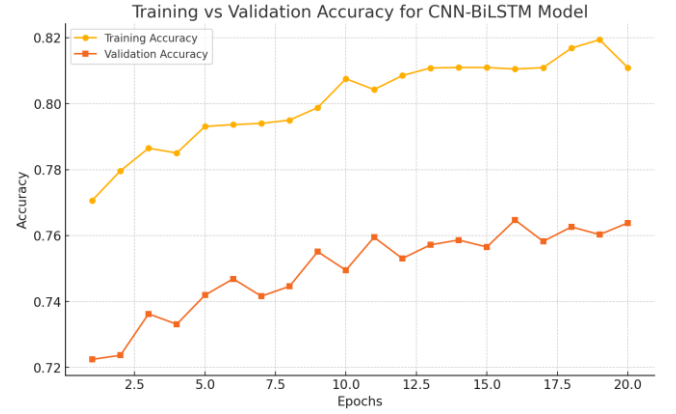


Figure 5. Training vs Validation Accuracy for CNN-BiLSTM Model

System Goals and Impact

The goal of the proposed system is twofold. The twofold, consisting of (1) dialect detection and (2) dialect conversion. The system integrates multiple layers that work together to process speech input and generate accurate, dialect-specific output.

Speech Processing and Feature Extraction (First Fold)

In the first stage, the input audio signal $s(t)$ is pre-processed and segmented into short frames. Each frame is transformed into its spectral representation using techniques such as the Short-Time Fourier Transform (STFT). From this, relevant acoustic features—including Mel-Frequency Cepstral Coefficients (MFCCs), pitch, and energy—are extracted to capture dialectal variations.

Mathematically, the framed signal is represented as:

$$X(k, n) = \sum_{m=0}^{N-1} s(m + nH) \cdot w(m) \cdot e^{-j2\pi km/N}$$

where:

$w(m)$ is the window function,

H is the hop size,

N is the frame length.

These extracted features form the basis for classification.

Dialect Classification and Conversion (Second Fold)

The second stage involves dialect detection followed by dialect conversion.

The extracted features are passed to a machine-learning model or AI service (e.g., Gemini AI), which classifies the input into its corresponding Kannada dialect using probability scores:

$$\hat{d} = \arg \max_{d \in D} P(d | X)$$

Once the dialect is identified, the text is converted into the target dialect using rule-based mapping or neural text generation techniques. Finally, the converted text is synthesized into audio using a Text-to-Speech (TTS) engine such as gTTS.

Technological Advancement: Improve dialect-specific speech recognition and language processing tools for Kannada [7][13][15].

Cultural Preservation: Support linguistic documentation and educational localization, ensuring dialects like Bangalore, Dharwad, and Brahmin are digitally preserved [19][20].

By integrating AI services such as Gemini AI [8] with NLP and deep learning techniques [7][15][16], the Kannada Dialect Detector and Converter bridges a major gap in regional language processing. This innovation not only enhances practical applications like voice assistants, transcription systems, and e-learning platforms, but also contributes to the preservation of Karnataka's linguistic heritage.

V. CONCLUSION AND FUTURE WORK

The Kannada Dialect Detector and converter initiative has achieved notable progress in creating a tool capable of accurately identifying and analyzing the diverse dialects of the Kannada language. Among its key achievements is the development of a powerful machine learning model that utilizes advanced speech recognition and multiple techniques for processing natural language to differentiate between dialects. Throughout the project, Important lessons were discovered regarding about the complexities of dialectal variations and The import of reliable data collection to ensure comprehensive representation. Expanding the data model to include underrepresented dialects remains a vital future step, as it will enhance The versatility of the model and perform effectively across various contexts. Additionally, incorporating user feedback mechanisms can facilitate real-time system improvements and better align the tool with user needs.

Deployment strategies are a critical focus for the project's next phase. Transitioning the prototype into a fully functional web service will need meticulous preparation, including, with considerations for scalability and user accessibility. Developing a mobile application version will further broaden user engagement, enabling on-the-go access to dialect detection capabilities. These efforts aim to make the tool more practical and widely available, supporting both individual users and broader applications such as education, linguistics, and content localization.

The project's impact extends beyond technological innovation to cultural preservation. By addressing linguistic nuances, the Kannada Dialect Detector not only enhances speech-driven applications like virtual assistants and transcription tools but additionally adds to the appreciation and preservation of Kannada's rich linguistic heritage. This project serves as a model for future research in dialect detection for other languages, fostering inclusivity in technology-driven solutions and bridging gaps in current voice recognition software for regional languages. Through the incorporation of cultural and technological advancements, the Kannada Dialect Detector and converter lays the groundwork for innovative solutions designed to meet the requirements of diverse linguistic communities

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